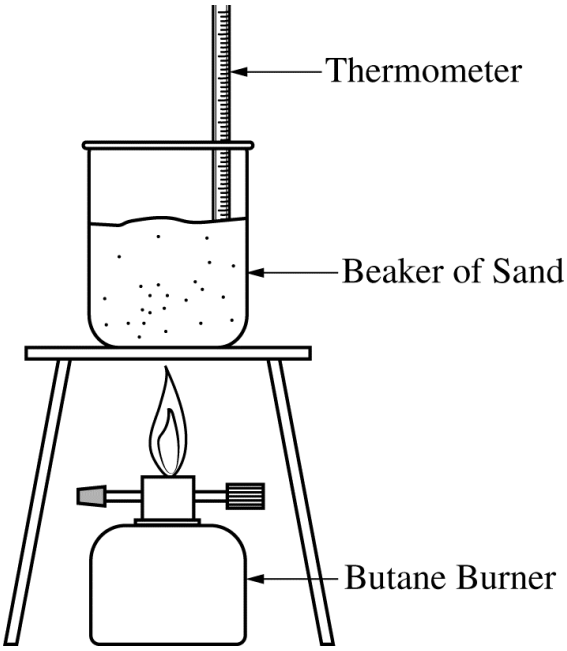
frq1

1. A student is asked to determine what mass of butane, $C_4H_{10}(g)$, needs to burn in order to raise the temperature of a 1650 g beaker of sand by 180.°C. The student is provided with the equipment shown below.



a. Calculate the amount of heat energy needed to raise the temperature of the beaker of sand by 180.°C. Assume that all the heat energy from the burner is transferred to the beaker of sand and that the specific heat capacity of the beaker and sand together is 0.810 J/(g×°C).

The student runs the experiment and collects the data shown in the table below.

Mass of the beaker and sand	1650 g
Mass of butane burner before combustion	225.26 g
Mass of butane burner after combustion	218.20 g
Initial temperature of the beaker of sand	20.°C
Final temperature of the beaker of sand	200.°C

b. Calculate the number of moles of butane that was used in the experiment. Report your answer to the appropriate number of significant figures.

The combustion of butane is represented by the equation below.

$$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 5H_2O(g) + 4CO_2(g) \Delta H_{rm}^{\circ} = -2659 \text{kJ/mol}_{rxn}$$

c. Using the balanced equation for the combustion of butane and $\Delta H_{rxn}°$, determine the amount of heat energy produced by the combustion of butane in the experiment.

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d. The student claims that some of the heat energy produced by the combustion of butane was lost to the air surrounding the system. Do your answers to parts (a) and (c) support the student's claim? Explain.

Part A

1 point is earned for the correct calculation of the energy.

$$q = mc\Delta T$$
$$q = (1650\text{g})\left(0.810\frac{\text{J}}{\text{g}\cdot{}^\circ\text{C}}\right)(180\cdot{}^\circ\text{C}) = 241,000\text{J} = 241\text{kJ}$$

0	1 ✓

Student earns 1/1 point(s)

1 point is earned for the correct calculation of the energy.

$$q = mc\Delta T$$
$$q = (1650\text{g})\left(0.810\frac{\text{J}}{\text{g}\cdot{}^\circ\text{C}}\right)(180\cdot{}^\circ\text{C}) = 241,000\text{J} = 241\text{kJ}$$

Part B

1 point is earned for the correct calculation of the number of moles with the correct number of significant figures.

$$225.26\text{g} - 218.20\text{g} = 7.06\text{g butane}$$
$$7.06\text{g} \times \frac{1\text{mol}}{58.12\text{g butane}} = 0.121\text{mol butane}$$

0	1 ✓

Student earns 1/1 point(s)

1 point is earned for the correct calculation of the number of moles with the correct number of significant figures.

$$225.26\text{g} - 218.20\text{g} = 7.06\text{g butane}$$
$$7.06\text{g} \times \frac{1\text{mol}}{58.12\text{g butane}} = 0.121\text{mol butane}$$

Part C

1 point is earned for the correct determination of the amount of heat produced.

$$0.121\text{ mol butane} \times \frac{2659\text{kJ produced}}{1\text{ mol butane}} = 322\text{kJ}$$

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0	1 ✓

Student earns 1/1 point(s)

1 point is earned for the correct determination of the amount of heat produced.

$$0.121 \text{ mol butane } \times \frac{2659\text{kJ produced}}{1 \text{ mol butane}} = 322\text{kJ}$$

Part D

1 point is earned for a correct choice and a valid explanation.

> Yes, the answers to parts (a) and (c) support the hypothesis. The amount of heat generated from the combustion of the butane is greater than the amount of heat required to cause the temperature change of the sand, indicating that some of the heat from the combustion of butane was lost.

0	1 ✓

Student earns 1/1 point(s)

1 point is earned for a correct choice and a valid explanation.

> Yes, the answers to parts (a) and (c) support the hypothesis. The amount of heat generated from the combustion of the butane is greater than the amount of heat required to cause the temperature change of the sand, indicating that some of the heat from the combustion of butane was lost.

frq1

2. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about atomic structure and bonding.

(a) Write the complete ground-state electron configuration of the Kr atom.

(b) The Rb^{+} ion has the same electron configuration as the neutral Kr atom. Which has the larger radius, the Rb^{+} ion or the Kr atom? Justify your answer in terms of Coulomb's law.

(c) The composition of a sample of solid Rb containing a high percentage of the radioactive isotope $\mathrm{Rb}\text{-}83$ is given in the following table.

Isotope	Isotopic Mass (amu)	Percentage of Sample	Isotope Half-Life
$\mathrm{Rb}\text{-}83$	82.92	32.5%	86 days
$\mathrm{Rb}\text{-}85$	84.91	50.8%	stable
$\mathrm{Rb}\text{-}87$	86.91	16.7%	stable

(i) Using the values in the table, calculate the average atomic mass, in amu, of Rb in the sample.

(ii) $\mathrm{Rb}\text{-}83$ undergoes radioactive decay into gaseous $\mathrm{Kr}\text{-}83$, which leaves the solid sample. Over time, how will the average atomic mass of the remaining sample compare to the value calculated in part (c)(i) ?

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides a correct answer.

- $1s^2 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6$

OR

frq1

- $[\mathrm{Ar}]\ 4s^2\ 3d^{10}\ 4p^6$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct answer and a valid justification.

- The Kr atom. The outermost electrons of and Kr occupy the same shell Rb^+, but Kr has one less proton, resulting in a weaker Coulombic attraction between the Kr nucleus and its electrons; therefore, Kr has a larger radius than Rb^+.

Part C(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct calculated value.

- $0.325(82.92) + 0.508(84.91) + 0.167(86.91) = 84.6\ \mathrm{amu}$

Part C(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct answer.

- The average atomic mass will increase over time.

frq1

3. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to the analysis of $CaBr_2$.

(a) A student has a 10.0 g sample of $CaBr_2$. Show the setup of the calculation to determine the number of moles of $CaBr_2$ in the sample. Include units in the setup. (You do not need to do any calculations.)

(b) What number, in addition to the answer to part (a), is needed to determine the number of atoms of Ca in the sample?

(c) A different student is given a 10.0 g sample labeled $CaBr_2$ that may contain an inert (nonreacting) impurity. Identify a quantity from the results of laboratory analysis that the student could use to determine whether the sample was pure.

(d) Explain why $CaCl_2$ is likely to have properties similar to those of $CaBr_2$.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response gives one of the following setups (or an equivalent variant):

- [ ] $10.0 \text{ g } CaBr_2 \times \frac{1 \text{ mol } CaBr_2}{199.88 \text{ g } CaBr_2} = \text{number of moles of } CaBr_2$ (right side of equation not necessary)
- [ ] $\frac{1 \text{ mol } CaBr_2}{199.88 \text{ g}} = \frac{x \text{ mol } CaBr_2}{10.0 \text{ g}}$
- [ ] $\frac{10.0 \text{ g } CaBr_2}{199.88 \text{ g } CaBr_2/\text{mol}}$

  Note: Molar mass can be written with any number of significant figures.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response gives the term "Avogadro's number" or the value of Avogadro's number, 6.022×10^{23} (with any number

frq1

of significant figures, units of mol^{-1} or per mole are not required).

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response does give an appropriate result (such as one of the following):

- [ ] Mass of Ca in sample
- [ ] Number of moles of Ca in sample
- [ ] Mass of Br in sample
- [ ] Number of moles of Br in sample
- [ ] Mass or number of moles of element other than Ca or Br in sample

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that Cl is in the same group/family/column of the periodic table as Br OR that a Cl atom has the same number of valence electrons as a Br atom.

frq1

4. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is asked to prepare a buffer solution made with equimolar amounts of $CH_3NH_2(aq)$ and $CH_3NH_3Cl(s)$. The student uses $25.00\ \text{mL}$ of $0.100\ M\ CH_3NH_2(aq)$, which contains $0.00250\ \text{mol}$ of CH_3NH_2, to make the buffer.

(a) Calculate the mass of $CH_3NH_3Cl(s)$ that contains $0.00250\ \text{mol}$ of CH_3NH_3Cl.

The student has the following materials and equipment available.

• Distilled water	• Electronic balance	• $50\ \text{mL}$ beaker	• Pipets
• $0.100\ M\ CH_3NH_2(aq)$	• Weighing paper	• $10.0\ \text{mL}$ graduated cylinder	• pH meter
• Solid CH_3NH_3Cl	• $50.00\ \text{mL}$ buret	• Small spatula	

(b) The following table contains a partial procedure for making the buffer solution. Fill in steps 1 and 4 to complete the procedure using only materials and equipment selected from the choices given. (Not all materials listed will be used. Assume that all appropriate safety measures are already in place.)

frq1

Step	Procedure
1	
2	Place the solid in the 50mL beaker.
3	Clean the buret and rinse with distilled water.
4	
5	Use the buret to add 25.00mL of $0.100M\text{CH}_3\text{NH}_2(aq)$ to the beaker.
6	Mix well.
7	Check the pH with the pH meter.

The value of K_b for $\text{CH}_3\text{NH}_2(aq)$ is 4.4×10^{-4}, and the pH of the buffer the student prepared is 10.64.

(c) The student prepares a second buffer solution. The student uses $25.00\ \text{mL}$ of $0.050\ M\ \text{CH}_3\text{NH}_2(aq)$ instead of $25.00\ \text{mL}$ of $0.100\ M\ \text{CH}_3\text{NH}_2(aq)$, and half the mass of $\text{CH}_3\text{NH}_3\text{Cl}(s)$ that was used in the first buffer. Is the pH of the second buffer greater than, less than, or equal to the pH of the first buffer? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct calculated value.

- $0.00250\text{ mol CH}_3\text{NH}_3\text{Cl} \times \frac{67.52\text{ g}}{1\text{ mol}} = 0.169\text{ g}$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

frq1

0	1	2 ✓

The response meets **both** of the following criteria.

- ☐ The response provides a correct description of step 1.
  - Use the spatula, balance, and weighing paper to measure out exactly 0.169 g of $CH_3NH_3Cl(s)$.

    OR

  - Use the balance to weigh out the mass of solid in part (a).
- ☐ The response provides a correct description of step 4.
  - Rinse the buret with a small amount of $0.100\ M\ CH_3NH_2(aq)$, drain, and refill with $0.100\ M\ CH_3NH_2(aq)$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

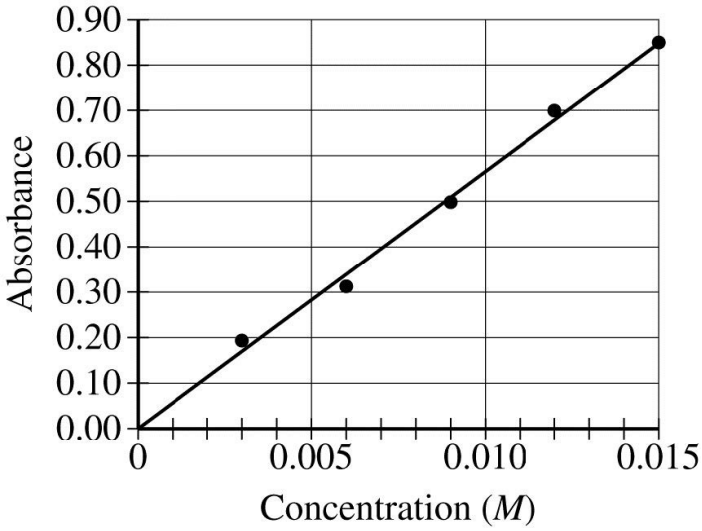
0	1 ✓

The response provides the correct answer and a valid justification.

- Equal to. The ratio of weak acid to conjugate base is still $1:1$.

frq1

5. To spectrophotometrically determine the mass percent of cobalt in an ore containing cobalt and some inert materials, solutions with known $[Co^{2+}]$ are prepared and the absorbance of each of the solutions is measured at the wavelength of optimum absorbance. The data are used to create a calibration plot, shown below.



A 0.630 g sample of the ore is completely dissolved in concentrated $HNO_3(aq)$. The mixture is diluted with water to a final volume of 50.00 mL. Assume that all the cobalt in the ore sample is converted to $Co^{2+}(aq)$.

a. What is the $[Co^{2+}]$ in the solution if the absorbance of a sample of the solution is 0.74?

b. Calculate the number of moles of $Co^{2+}(aq)$ in the 50.00 mL solution.

c. Calculate the mass percent of Co in the 0.630 g sample of the ore.

Part A

1 point is earned for a concentration that is consistent with an absorbance of 0.74 on the graph.

An absorbance of 0.74 corresponds to a concentration of 0.0130 *M* (0.0125 *M* to 0.0135 *M*).

0	1 ✓

Student earns 1/1 point(s)

1 point is earned for a concentration that is consistent with an absorbance of 0.74 on the graph.

An absorbance of 0.74 corresponds to a concentration of 0.0130 *M* (0.0125 *M* to 0.0135 *M*).

Part B

1 point is earned for the correct calculation of the number of moles.

frq1

$$0.05000\text{L} \times \frac{0.0130\text{mol}\text{Co}^{2+}}{1.00\text{L}} = 6.50 \times 10^{-4}\text{mol}\text{Co}^{2+}$$

0	1 ✓

Student earns 1/1 point(s)

1 point is earned for the correct calculation of the number of moles.

$$0.05000\text{L} \times \frac{0.0130\text{mol}\text{Co}^{2+}}{1.00\text{L}} = 6.50 \times 10^{-4}\text{mol}\text{Co}^{2+}$$

Part C

1 point is earned for a correct calculation of the number of grams of Co that is consistent with the number of moles of Co2+ in part (b). (may be implicit)

1 point is earned for a correct calculation of the mass percent that is consistent

with the number of grams of Co.

$$6.50 \times 10^{-4}\text{mol} \times \frac{58.93\text{gCo}}{1.00\text{mol}} = 0.0383\text{gCo}$$
$$\frac{0.0383\text{gCo}}{0.630\text{g ore}} \times 100\% = 6.08\%\text{Co}$$

0	1	2 ✓

Student earns 2/2 point(s)

1 point is earned for a correct calculation of the number of grams of Co that is consistent with the number of moles of Co2+ in part (b). (may be implicit)

1 point is earned for a correct calculation of the mass percent that is consistent

with the number of grams of Co.

$$6.50 \times 10^{-4}\text{mol} \times \frac{58.93\text{gCo}}{1.00\text{mol}} = 0.0383\text{gCo}$$
$$\frac{0.0383\text{gCo}}{0.630\text{g ore}} \times 100\% = 6.08\%\text{Co}$$

frq1

6. **For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**

$$4\,\mathrm{Fe}(s) + 3\,\mathrm{O_2}(g) \rightarrow 2\,\mathrm{Fe_2O_3}(s) \qquad \Delta H^\circ = -1650\,\mathrm{kJ/mol}_{rxn}$$

A student investigates a reaction used in hand warmers, represented above. The student mixes $\mathrm{Fe}(s)$ with a catalyst and sand in a small open container. The student measures the temperature of the mixture as the reaction proceeds. The data are given in the following table.

Time (min)	Temperature of Mixture$(^\circ\mathrm{C})$
0	22.0
1	25.1
2	34.6
3	37.3
4	39.7
5	39.4

(a) The mixture ($\mathrm{Fe}(s)$, catalyst, and sand) has a total mass of 15.0 g and a specific heat capacity of $0.72\ \mathrm{J/(g \cdot {}^\circ C)}$. Calculate the amount of heat absorbed by the mixture from 0 minutes to 4 minutes.

(b) Calculate the mass of $\mathrm{Fe}(s)$, in grams, that reacted to generate the amount of heat calculated in part (a).

(c) In a second experiment, the student uses twice the mass of iron as that calculated in part (b) but the same mass of sand as in the first experiment. Would the maximum temperature reached in the second experiment be greater than, less than, or equal to the maximum temperature in the first experiment? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response does not provide the correct calculated value with units:

$\cdot\, q = mc\Delta T = (15.0\,\mathrm{g})(0.72\,\mathrm{J/(g \cdot {}^\circ C)})(39.7^\circ\mathrm{C} - 22.0^\circ\mathrm{C}) = 190\,\mathrm{J}$

Part B

frq1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response meets **all** the following criteria.

- ☐ The response provides the correct calculated value of the moles of reaction, consistent with part (a) (may be implicit).
  $q_{sys} = -q_{surr}$
  $-190 \text{ J} \times \frac{1 \text{ kJ}}{1000 \text{ J}} \times \frac{1 \text{ mol}_{rxn}}{-1650 \text{ kJ}} = 0.00012 \text{ mol}_{rxn}$
- ☐ The response provides the correct calculated value of the mass of iron.
  $0.00012 \text{ mol}_{rxn} \times \frac{4 \text{ mol Fe}}{1 \text{ mol}_{rxn}} \times \frac{55.85 \text{ g Fe}}{1 \text{ mol Fe}} = 0.027 \text{ g Fe}$
  ($0.026\ g$ if decimals are carried)

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

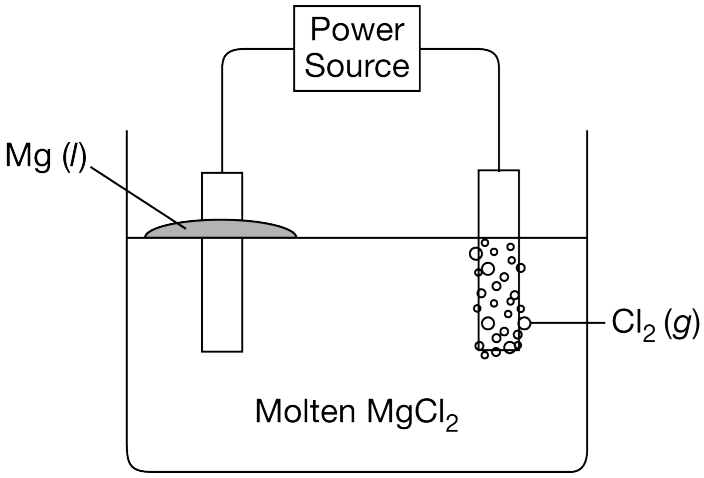
0	1 ✓

The response provides a correct answer and a valid justification.

Greater than. A greater mass of iron provides a greater number of moles of reaction, which would transfer a greater quantity of thermal energy to the same mass of sand and therefore lead to a greater maximum temperature.

frq1

7. **For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.**



Half-Reaction	$E°$ (V)
$Mg^{2+} + 2\,e^- \rightarrow Mg$	−2.37
$Cl_2 + 2\,e^- \rightarrow 2\,Cl^-$	+ 1.36

Molten $MgCl_2$ can be decomposed into its elements if a sufficient voltage is applied using inert electrodes. The products of the reaction are liquid Mg (at the cathode) and Cl_2 gas (at the anode). A simplified representation of the cell is shown above. The reduction half-reactions related to the overall reaction in the cell are given in the table.

(a) Draw an arrow on the diagram to show the direction of electron flow through the external circuit as the cell operates.

(b) Would an applied voltage of $2.0\ \text{V}$ be sufficient for the reaction to occur? Support your claim with a calculation as part of your answer.

(c) If the current in the cell is kept at a constant 5.00 amps, how many seconds does it take to produce $2.00\ \text{g}$ of $Mg(l)$ at the cathode?

Part A

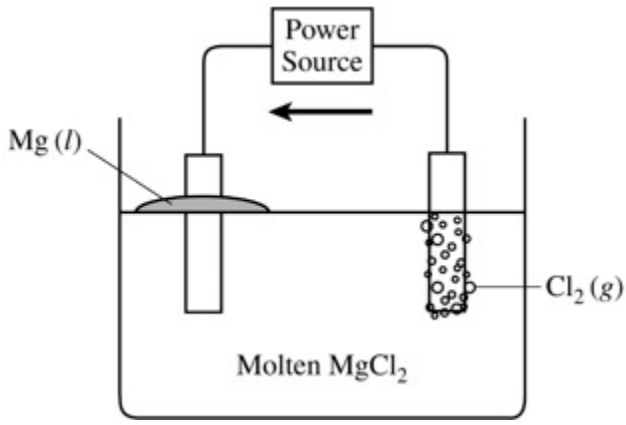
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct answer.

· Electron flow should be indicated only in a counter-clockwise direction in the external circuit, from the Cl_2 anode to the Mg cathode.

frq1



Parrt B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct answer and a calculated value.

· No, because $2.0\ \mathrm{V}$ is less than $3.73\ \mathrm{V}$, which is the minimum voltage needed for electrolysis to occur.

$\mathrm{E}^{\circ}_{\mathrm{cell}} = -2.37\ \mathrm{V} + (-1.36\ \mathrm{V}) = -3.73\ \mathrm{V}$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response meets **all** the following criteria.

- [ ] The response provides the correct calculated value of moles of electrons (may be implicit).
  · $2.00\ \mathrm{g\ Mg} \times \frac{1\ \mathrm{mol\ Mg}}{24.30\ \mathrm{g\ Mg}} \times \frac{2\ \mathrm{mol}\ e^-}{1\ \mathrm{mol\ Mg}} = 0.165\ \mathrm{mol}\ e^-$
- [ ] The response provides the correct calculated number of seconds.
  · $0.165\ \mathrm{mol}\ e^- \times \frac{96{,}485\ \mathrm{C}}{1\ \mathrm{mol}\ e^-} \times \frac{1\ \mathrm{s}}{5.00\ \mathrm{C}} = 3180\ \mathrm{s}$

frq1

8. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

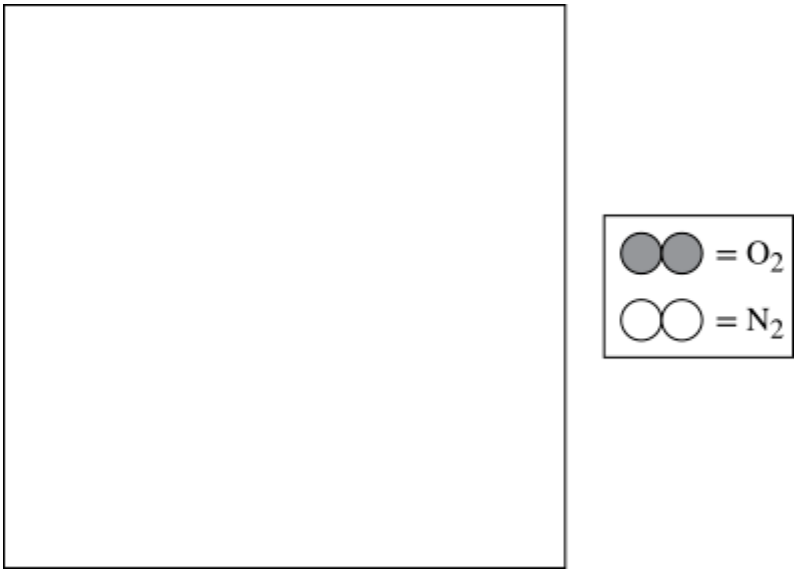
Dry air is comprised of N_2, O_2, and traces of other gases.

A student obtains a $2.00\ L$ container filled with a sample of dry air at a total pressure of $1.30\ atm$ and a temperature of $295\ K$.

(a) Calculate the total number of moles of gas in the container.

(b) The student cools the gaseous mixture until either N_2 or O_2 condenses. Which gas will condense first? Justify your answer in terms of the types and relative strengths of the intermolecular forces in N_2 and O_2.

(c) Using the legend given, draw at least three molecules of each type of gas in the sealed container represented below when one of the substances has condensed, but the other substance is still a gas.



(d) In this sample, 78% of the molecules are N_2 and 21% of the molecules are O_2. Calculate the total number of grams of the substance that condenses first.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

frq1

0	1 ✓

The response provides the correct calculated value.

- $PV = nRT$

  $n = \frac{PV}{RT} = \frac{(1.30\text{ atm})(2.00\text{ L})}{(0.08206\text{ L atm mol}^{-1}\text{ K}^{-1})(295\text{ K})} = 0.107\text{ mol}$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct answer and a valid justification.

- O_2 will condense first. Both N_2 and O_2 have only dispersion forces as their intermolecular forces. Because O_2 has more electrons, it has a higher polarizability. Therefore, O_2 will have stronger dispersion forces and thus will condense at the higher temperature.

Part C

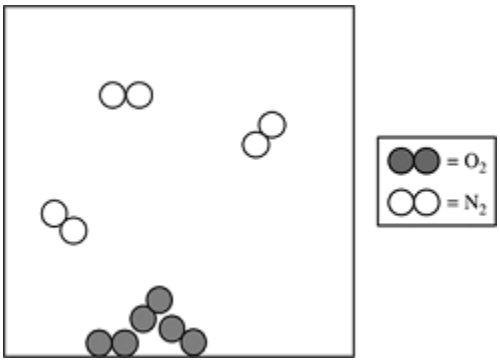
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides a correct diagram, consistent with part (b).

- Diagram should show one substance is a gas and the other substance is a liquid.

frq1



Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

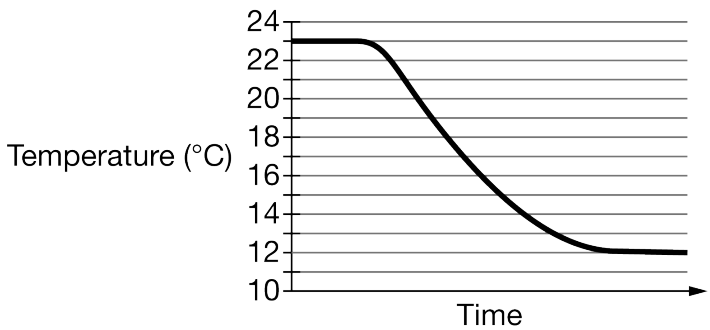
0	1 ✓

The response provides the correct calculated value, consistent with part (b).

- $(0.21)(0.107\text{ mol})(32.00\text{ g/mol}) = 0.72\text{ g }O_2$

frq1

9. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student conducts an experiment to determine the value of ΔH°_{soln} for the dissolution of $NaC_2H_3O_2(s)$. The student dissolves 10.0 g of $NaC_2H_3O_2(s)$ in room-temperature water in a beaker and measures the temperature over time. The data are given in the graph above.

(a) The student touches the side of the beaker after the dissolution has occurred and observes that it is cold. What experimental evidence is consistent with the student's observation?

(b) Is the dissolution of $NaC_2H_3O_2(s)$ endothermic or exothermic? Justify your answer in terms of the flow of energy between the system and surroundings.

(c) The student calculated the energy change for the dissolution to be 4600 J. Based on this value, calculate ΔH°_{soln} in kJ/mol.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides a correct answer.

· According to the graph, the temperature of the mixture after dissolution is lower, which is consistent with the observation that the beaker feels cold after the dissolution.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

frq1

0	1 ✓

The response provides the correct answer and a valid justification.

· Endothermic. The decrease in temperature indicates that energy flows from the beaker and water (surroundings) to the dissolving solid (system).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

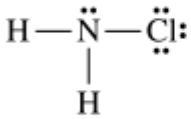
The response meets **all** the following criteria.

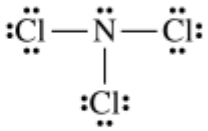
- ☐ The response provides the correct calculated number of moles (may be implicit):
  · $10.0 \text{ g } NaC_2H_3O_2 \times \frac{1 \text{ mol}}{82.03 \text{ g}} = 0.122 \text{ mol } NaC_2H_3O_2$
- ☐ The response provides the correct calculated enthalpy change (with units) consistent with the number of moles.
  · $\Delta H^\circ_{soln} = \frac{4600 \text{ J}}{0.122 \text{ mol}} = 38,000 \text{ J/mol} = 38 \text{ kJ/mol}$

frq1

10. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about the compounds NH_2Cl and NCl_3. The Lewis electron-dot diagrams of the two compounds are shown.





(a) Calculate the number of moles of NH_2Cl (molar mass $51.48\ \text{g/mol}$) present in $1.0\ \text{L}$ of a solution in which the concentration of NH_2Cl is $0.0016\ \text{g/L}$.

(b) NH_2Cl is highly soluble in water, whereas NCl_3 is nearly insoluble. Explain this observation in terms of the types and relative strengths of the intermolecular forces between each of the solutes and water.

(c) The value of $\Delta H^\circ_{vaporization}$ for $NCl_3(l)$ is $32.9\ \text{kJ/mol}$. Calculate the amount of energy required to vaporize a $15.0\ \text{g}$ sample of NCl_3 (molar mass $120.36\ \text{g/mol}$).

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct calculated value.

- $1\,\text{L} \times \frac{0.0016\,\text{g}}{1\,\text{L}} \times \frac{1\,\text{mol}}{51.48\,\text{g}} = 3.1 \times 10^{-5}\ \text{mol}$

Part B

frq1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response meets **both** of the following criteria.

- ☐ The response provides the correct identification of intermolecular forces between each substance and water.
  - Both NH_2Cl and NCl_3 can participate in hydrogen bonding with water.

    OR

  - Both NH_2Cl and NCl_3 have dipole-dipole attractions to water.
- ☐ The response provides a correct explanation.
  - The intermolecular forces between NH_2Cl molecules and water are stronger than those between NCl_3 molecules and water, which leads to the greater solubility of NH_2Cl in water.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

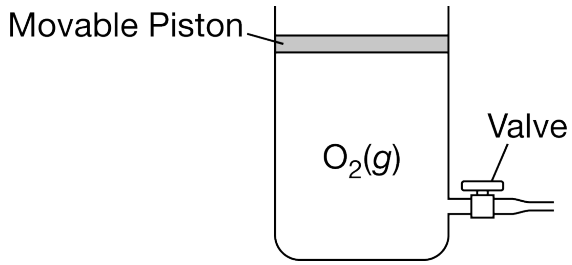
0	1 ✓

The response provides the correct calculated value.

- $15.0\,\text{g}\,NCl_3 \times \frac{1\,\text{mol}}{120.36\,\text{g}} \times \frac{32.9\,\text{kJ}}{1\,\text{mol}} = 4.10\,\text{kJ}$

frq1

11. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student investigates gas behavior using a rigid cylinder with a movable piston of negligible mass, as shown in the diagram above. The cylinder contains $0.325\ \text{mol}$ of $O_2(g)$.

(a) The cylinder has a volume of $7.95\ \text{L}$ at 25°C and $1.00\ \text{atm}$. Calculate the density of the $O_2(g)$, in g/L, under these conditions.

(b) Attempting to change the density of the $O_2(g)$, the student opens the valve on the side of the cylinder, pushes down on the piston to release some of the gas, and closes the valve again. The temperature of the gas remains constant at 25°C. Will this action change the density of the gas remaining in the cylinder? Justify your answer.

(c) The student tries to change the density of the $O_2(g)$ by cooling the cylinder to -55°C, which causes the volume of the gas to decrease. Using principles of kinetic molecular theory, explain why the volume of the $O_2(g)$ decreases when the temperature decreases to -55°C.

(d) The student further cools the cylinder to -180°C and observes that the measured volume of the $O_2(g)$ is substantially smaller than the volume that is calculated using the ideal gas law. Assume all equipment is functioning properly. Explain why the measured volume of the $O_2(g)$ is smaller than the calculated volume. (The boiling point of $O_2(l)$ is -183°C.)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct calculated value.

$$0.325\ \text{mol O}_2 \times \frac{32.00\ \text{g O}_2}{1\ \text{mol O}_2} = 10.4\ \text{g O}_2$$

$$D = \frac{m}{V} = \frac{10.4\ \text{g}}{7.95\ \text{L}} = 1.31\ \text{g/L}$$

OR

frq1

$\cdot D = \frac{m}{V} = \frac{P(MM)}{RT} = \frac{(1.0\text{ atm})(32.00\text{ g/mol})}{(0.08206\frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}})(298\text{ K})} = 1.31\text{ g/L}$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides a correct answer and a valid justification.

· No, the density of the gas remains constant because P, R, and T remain constant AND the mass and volume of O_2 decrease proportionately.

OR

· A mathematical justification is shown below.

$$D = \frac{m}{V} = \frac{n\text{ moles of }O_2 \times \text{molar mass of }O_2}{\frac{nRT}{P}} = \frac{P\times(\text{molar mass of }O_2)}{RT}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides a valid explanation.

· As the gas cools, the average kinetic energy (speed) of the O_2 molecules decreases. The molecules rebound with less energy when they collide with each other and the walls of the container. The spacing between particles decreases, causing the volume occupied by the gas to decrease.

OR

· As the gas cools, the average kinetic energy (speed) of the O_2 molecules decreases. The molecules rebound with less energy when they collide with each other and the walls of the container. The only way for the molecules to maintain a constant rate of collisions with the walls of the container (maintaining a pressure of 1.00 atm) is for the volume of the gas to decrease.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

frq1

0	1 ✓

The response provides a valid explanation.

· The ideal gas law assumes that gas particles do not experience interparticle attractions. As a real gas cools further, the intermolecular forces have greater effect as the average speed of the molecules decreases, resulting in inelastic collisions. To maintain a gas pressure of $1.00\ \mathrm{atm}$, the volume must decrease to accommodate more collisions with less energy.

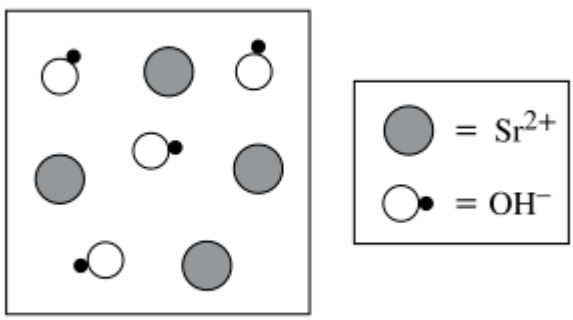
frq1

12. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Strontium hydroxide dissolves in water according to the following equation. The K_{sp} expression for strontium hydroxide is provided.

$$\mathrm{Sr(OH)_2}(s) \rightleftarrows \mathrm{Sr^{2+}}(aq) + 2\ \mathrm{OH^-}(aq) \qquad K_{sp} = \left[\mathrm{Sr^{2+}}\right]\left[\mathrm{OH^-}\right]^2$$



(a) A student draws the particulate diagram shown to represent the ions present in an aqueous solution of $\mathrm{Sr(OH)_2}$. (Water molecules are intentionally omitted.) Identify the error in the student's drawing.

(b) The student prepares a saturated solution by adding excess $\mathrm{Sr(OH)_2}(s)$ to distilled water and stirring until no more solid dissolves. The student then determines that $\left[\mathrm{Sr^{2+}}\right] = 0.043\ M$ in the solution.

(i) Calculate the value of $\left[\mathrm{OH^-}\right]$ in the solution.

(ii) Calculate the value of K_{sp} for $\mathrm{Sr(OH)_2}$.

(c) The student prepares a second saturated solution of $\mathrm{Sr(OH)_2}$ in aqueous $0.10\ M\ \mathrm{Sr(NO_3)_2}$ instead of water. Will the value of $\left[\mathrm{OH^-}\right]$ in the second solution be greater than, less than, or equal to the value in the first solution? Justify your answer. (Assume constant temperature.)

Part A

frq1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides a correct answer.

- The student's drawing shows an incorrect ratio of Sr^{2+} and OH^- ions.

  OR

- The student's drawing is not charge-balanced.

Part B(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct calculated value.

- $\frac{0.043 \text{ mol } Sr^{2+}}{1 \text{ L}} \times \frac{2 \text{ mol } OH^-}{1 \text{ mol } Sr^{2+}} = 0.086\ M\ OH^-$

Part B(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response provides the correct calculated value, consistent with part (b)(i).

- $K_{sp} = [Sr^{2+}][OH^-]^2 = (0.043)(0.086)^2 = 3.2 \times 10^{-4}$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

frq1

The response provides the correct answer and a valid justification.

- Less than. Because the $Sr(NO_3)_2(aq)$ solution already contains a common ion, $Sr^{2+}(aq)$, the solubility of $Sr(OH)_2$ will be decreased, resulting in a lower value of $[OH^-]$.